

Mechanical Dynamics & Enclosure IP Validation Test Report

Vibration Shaker Endurance and Water Ingress per ISO 16750-3 / IEC 60529

1. 3-Axis Random Vibration Shaker Test

- Test Case TC-MECH-093: Random vibration test completed on X-axis (8.0 h) and Y-axis (8.0 h) at 2.50 g RMS with zero mechanical failure.
- Frequency spectrum: 10 Hz to 2000 Hz broadband random excitation profile.

2. Enclosure Ingress Protection (IP67) Submersion Test

- Test Case TC-MECH-095: Ingress protection test: housing passed IP6X dust test, but water ingress of 4.2 mL observed during 1m submersion.